Informative dropout and trial-design choice in aggregated N-of-1 biomarker-validation trials: a paired simulation at one parameter point

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Abstract

Background. Participants leave aggregated N-of-1 trials, and their reasons can depend on how their symptoms change. We asked how dropout affects the test of a biomarker-by-treatment interaction in three designs.

Methods. We simulated open-label titration with blinded discontinuation (OL+BDC), a two-period crossover (CO) and the Hybrid design at one parameter point, with $N = 70$ in each. The response and baseline parameters were those of the published code of Hendrickson et al. The biomarker shifted the drug-response mean at on-drug visits. The true interaction was -3.60 points per biomarker standard deviation. Seven dropout conditions were applied to the same 1,000 complete datasets per design. They were no dropout, completely random dropout and magnitude-driven dropout at 25 and 40 percent, and three directional mechanisms at 25 percent. We fitted the mixed model at analysis half-lives of 0.5 week and 0, with model-based and cluster-robust (CR2) tests. A null regime had 1,000 datasets per design.

Results. The 0.5-week analysis half-life biased the interaction estimate by -0.34 in OL+BDC and -0.23 in the Hybrid design (MCSE 0.04). A binary exposure indicator, which matches the generating interaction, removed that bias. Without dropout the empirical standard deviation of the estimate was 1.120 (Hybrid), 1.139 (OL+BDC) and 1.376 (CO). Dropout at 25 percent

raised the standard error by a factor of 1.079 to 1.089 under completely random dropout. At 40 percent magnitude-driven dropout the factor was 1.118 to 1.154, nearly the same in all three designs. CO lost more absolute power than the other two designs in every condition. Its lower starting power did not explain the difference. Worsening-, improvement- and symptom-return-driven dropout attenuated the estimate in every design, most in CO under the binary indicator. Improvement-driven dropout attenuated it by 0.072 to 0.204 points (paired MCSE at most 0.021). Dropout did not change test size. In separate null checks the CR2 test rejected in 0.050 to 0.057 of datasets in the three designs. The model-based test was conservative in OL+BDC, rejecting in 0.035 of 5,000 null datasets (MCSE 0.003).

Conclusions. At this parameter point the Hybrid and OL+BDC designs carried the same information and lost it to dropout at the same rate. CO needed about 1.5 times the sample for the same precision. The analysis half-life must match the generating interaction. The rate of dropout set the loss of precision. Dropout that depended on the direction of symptom change attenuated the estimate, most in CO when the analysis half-life matched.

1 Introduction

1.1 The question

Aggregated N-of-1 trials pool many single-participant series of on-drug and off-drug periods into one mixed model[1]. A predictive-biomarker trial of this kind tests whether a baseline biomarker modifies the drug effect. The target is the biomarker-by-treatment interaction $\beta_{bm:D}$. Participants leave such trials, and their reasons can depend on how their symptoms are changing. We ask whether three candidate designs differ in how much power to detect $\beta_{bm:D}$ they lose to dropout, and whether the answer depends on the dropout mechanism or only on its rate.

The three designs are open-label titration followed by blinded discontinuation (OL+BDC), the traditional two-period crossover (CO), and the Hybrid design of Hendrickson et al.[2]. The Hybrid design is open-label titration, then blinded discontinuation, then a brief crossover. A fourth design, open label throughout (OL), has no off-drug visits. It cannot separate the drug response from the placebo response and the natural history, so it carries no information on $\beta_{bm:D}$. We exclude it from the comparison for that reason (Section 3).

1.2 Prior work

Hendrickson et al.[2] introduced the Hybrid design and compared it with OL, OL+BDC and CO by simulation. Their censoring filter assigned each simulated participant a per-unit-time dropout probability equal to a flat hazard plus a term in the square of the symptom change since baseline, shifted by 100. They describe this as a probability that is higher for participants who worsen or remain symptomatic and lower for participants who benefit (Methods). They report that increased censoring lowered power in every design, with a faster decline for OL+BDC because of participant loss before the blinded phase (Results, Figure

4A). In their comparison the Hybrid design had higher power than OL and OL+BDC and slightly lower power than CO without carryover. Its power fell faster than that of CO once carryover was introduced. Type I error rose with censoring, most for OL+BDC (Figure 6). Under the null, the interaction estimate was biased toward negative values for most censoring patterns in OL+BDC and for several in CO (Figure 7A). Against an uncensored reference, censoring changed the estimate only in the OL design (Figure 7B). Their 'happy accident' refers to trial cost. Participants with no response, or with a response clearly lost on placebo, tend to leave before the later crossover blocks, so those blocks are spent on participants whose earlier phases were inconclusive (Discussion). It is not a statement about power.

The present study does not reproduce those results. It uses their response and baseline parameters. It uses different correlation parameters, carryover, analysis model and dropout hazards (Section 7.2).

Matthews and Henderson[3] studied two-period crossover designs under non-ignorable dropout. They found the conventional AB/BA design robust under a missing-at-random analysis, and they identified a sequence set that is preferable when dropout depends on the within-participant difference between periods. Blackston et al.[4] compared aggregated N-of-1 designs with parallel and crossover trials by simulation, including selection bias. N-of-1 reporting guidance asks that dropout be reported[5], and reviews of published N-of-1 studies assess how completely trials report their methods[6,7]. We did not find a simulation that compares OL+BDC, CO and the Hybrid design on paired data under several dropout mechanisms.

The missing-data background is standard. Rubin[8] defined data missing completely at random (MCAR), missing at random (MAR) and missing not at random (MNAR)[9]. Selection models[10], pattern-mixture models[11,12] and shared-parameter models[13,14] describe dropout that depends on unobserved values. Likelihood-based mixed models are the usual primary analysis when dropout is plausibly MAR[15–19]. Sensitivity analyses vary the departure from MAR through delta-adjusted or reference-based multiple imputation[20–24] and tipping-point displays[25]. The ICH E9(R1) addendum treats dropout as an intercurrent event to be handled through the estimand[26,27]. This paper uses only the first of these tools, the likelihood-based mixed model. Joint models for longitudinal and event-time data[28,29] and run-in selection[30] are related but are not used here.

1.3 What this paper contributes

We make three contributions. First, we define the simulated dropout mechanisms exactly and state their missing-data properties (Section 2). Second, we report a paired simulation in which each dropout condition is applied to the same complete datasets. Power losses, coefficient shifts and standard-error ratios are then paired differences with small Monte Carlo standard errors (MCSE). Third, we show that the analysis half-life must match the generating interaction. The historical analysis decays the exposure indicator at 0.5 week, while the generating interaction has no carryover. This mismatch biases the estimate for OL+BDC and the Hybrid design but not for CO (Section 5.2).

The study covers one parameter point. It is exploratory. No analysis plan was registered in

advance, and its findings should be confirmed in an independent study. Three components of the working protocol were not carried out: the sample-size, effect-size and carryover sweeps; a path-stratified decomposition of the dropout pattern in the Hybrid design; and analysis-stage sensitivity analyses under alternative missing-data assumptions. Section 7.3 lists what each would add.

2 Dropout mechanisms and their missing-data properties

2.1 The simulated hazards

All dropout is generated by the function `first_dropout()` in the study driver (`analysis/scripts/informative`
The package function `censordata()` is not used (Section 2.2).

Let Y_{it} be the symptom score of participant i at post-baseline visit t and BL_i the baseline score. Higher scores are worse. The change from baseline is $\Delta_{it} = Y_{it} - \mathrm{BL}_i$, so a negative change is an improvement. The step change is $r_{it} = Y_{it} - Y_{i,t-1}$, with $Y_{i0} = \mathrm{BL}_i$. Within each randomization path and visit, let $\bar{\Delta}_t$ and s_t be the mean and standard deviation of Δ_{it} across the participants of that path, and define $\bar{r}_t$ and s_t^r in the same way. The raw hazard h_{it} of each condition is:

- MCAR: $h_{it} = 1$.
- Magnitude of change: $h_{it} = \min\{1, \beta_0 + \beta_1(\Delta_{it}/s_t)^2\}$ with $\beta_0 = 0.05$ and $\beta_1 = 0.5$. The change is scaled but not centered.
- Worsening-driven: $h_{it} = \exp\{\gamma(\Delta_{it} - \bar{\Delta}_t)/s_t\}$.
- Improvement-driven: $h_{it} = \exp\{-\gamma(\Delta_{it} - \bar{\Delta}_t)/s_t\}$.
- Symptom return: $h_{it} = \exp\{\gamma(r_{it} - \bar{r}_t)/s_t^r\}$.

We set $\gamma = \ln 3$. Under the three directional hazards the raw hazard triples for each standard deviation of the driving quantity.

The hazard applied at visit t is $\min(1, \alpha h_{it})$. The participant drops out at the first visit t with $U_{it} < \min(1, \alpha h_{it})$, where the U_{it} are independent uniform draws. That visit and all later visits are deleted. Baseline is always observed. The multiplier α is found by root finding on each realized path dataset. It solves

$$\frac{1}{n}\sum_i \Big[1 - \prod_t \{1 - \min(1, \alpha h_{it})\}\Big] = d,$$

where n is the number of participants on the path and d is the target fraction of participants who ever drop out (0.25 or 0.40). The achieved fraction therefore matches d in expectation by construction. It is not an outcome of the study.

The hazard is per visit, not per unit of time. Visits are 2.5 weeks apart throughout CO, 4 weeks apart in the open-label phases of OL+BDC and the Hybrid design, and 1 week apart

in their blinded phases. A common target d therefore places dropout at different calendar times and phases in different designs. Section 5.1 reports where dropout fell.

The 'biased' condition of the earlier unpaired prototype is the magnitude-of-change hazard. It rises with $|\Delta_{it}|$ in either direction. The mean change is an improvement at every post-baseline visit, so a large $|\Delta_{it}|$ is more often a large improvement than a large worsening. That condition is not worsening-driven, and we no longer label it so. The worsening-driven, improvement-driven and symptom-return hazards were added to separate the direction of selection from its magnitude.

2.2 Relation to `censordata()` and to the hazard of Hendrickson et al.

The package function `censordata()` combines a flat Bernoulli draw and a symptom-dependent Bernoulli draw. In the published code (commit 58b32a9) the symptom-dependent term is proportional to $(D + 100)^{e_{b1}}$, where D is the generated column `D_<visit>`. That column is the sum of the response components, which equals $\mathrm{BL}_i - Y_{it}$, the improvement, so the term increases with improvement. Hendrickson et al.[2] describe the term as the change in symptoms since baseline shifted by 100, which makes dropout higher for participants who worsen. The code and the description disagree in sign. The package and its live implementations now use $(100 - D)^{e_{b1}}$, as described; the frozen reference implementation keeps the published form. Both draws are proportional to the interval since the previous visit, `t_wk`, so the parameter β_0 is a per-visit hazard rather than a total dropout rate. The present study does not call `censordata()`, so none of these points affects its results.

2.3 Missing-data classification

Under the MCAR condition dropout is independent of all outcomes. It is missing completely at random[8].

Under the four outcome-dependent conditions, dropout at visit t depends on Y_{it}, and Y_{it} is then deleted. Dropout therefore depends on an unobserved value, and the data are MNAR. This is the selection-model form of Diggle and Kenward[10]. Two features weaken the practical consequence. First, the serial correlation of the generating process is $0.7^{|w_t - w_s|}$ within each response component, where w_t is the calendar week. Y_{it} is therefore partly predicted by $Y_{i,t-1}$, which is observed. We expect the mechanism to behave nearly as MAR, but we did not quantify this. Second, the magnitude-of-change hazard is symmetric in the direction of change. It removes strongly improving and strongly worsening participants alike.

MAR alone does not make the likelihood-based analysis unbiased. Valid likelihood inference under MAR also requires a correctly specified model for the full data[9,18]. The fitted mean is linear in calendar time with a random intercept and a continuous-time first-order autoregressive (CAR(1)) residual. The generating mean is a sum of Gompertz curves with three autocorrelated components. MAR dropout can therefore bias $\hat{\beta}_{bm:D}$ here, and any bias we observe cannot be attributed to MNAR alone.

We did not simulate shared-parameter dropout[13,14]. In that mechanism the hazard depends on a participant's latent drug response, which is the quantity the biomarker predicts. It is the mechanism most likely to bias $\beta_{bm:D}$ and the one that the clinical motivation most directly suggests. Our symptom-return hazard depends on the observed off-drug rise in symptoms. That rise is larger for participants with a larger drug response, so the hazard is related to the latent response only through the observed outcome.

2.4 Estimand

We treat dropout as an intercurrent event[26]. The target is the hypothetical-strategy estimand: $\beta_{bm:D}$ as it would be had every participant completed the schedule. Its true value is fixed by the generating process (Section 4).

3 The designs and where dropout falls

Each design is a set of randomization paths. A path is a sequence of visits, each on drug or off drug. The interaction is identified by within-participant contrasts between on-drug and off-drug visits. A dropout removes the remaining visits of a path, and with them any contrast they carry.

OL+BDC has eight visits at weeks 4, 8, 12, 16, 17, 18, 19 and 20. Both of its two paths are on drug through week 16. One path switches to placebo after week 18 and the other after week 17. The off-drug visits are therefore the last two or three, and a dropout before them removes the only off-drug information for that participant.

CO has eight visits 2.5 weeks apart. One path is on drug for the first four visits and on placebo for the last four. The other path is the reverse. Each participant contributes four visits on each side of the switch, and a dropout in the second period removes part of one side.

The Hybrid design has visits at weeks 4, 8, 9, 10, 11, 12, 16 and 20. All four paths are on drug through week 9. Two paths switch to placebo after week 9 and two after week 10. The last two visits, at weeks 16 and 20, are a drug-placebo crossover in either order. Off-drug visits occur in the blinded phase and in the crossover.

The OL design is on drug at every visit. The package analysis `lme_analysis()` detects that no participant varies in exposure after baseline and then fits `Sx ~ bm + t + bm:t`. The coefficient it tests is the biomarker-by-time slope, not $\beta_{bm:D}$. The earlier prototype, run under an earlier parameter set (Section 5.6), reported that slope test in its OL row. Its rejection rate, 0.086 to 0.112, was the power of a different test. It was neither a test size nor a measure of information on the interaction. A model with `bm:Dbc` and the baseline row would be estimable for OL, but its interaction is not separable from the natural-history and placebo components. We therefore exclude OL.

These descriptions say where each design's off-drug information lies. They are not tested hypotheses about which design loses most. The results in Section 5 address that question directly.

Table 1: Response and baseline parameters, read from the run metadata. They are the values of the published code of Hendrickson et al. Modified Gompertz maximum m (points), displacement d and rate r (per week). SD: standard deviation of the component, the biomarker or baseline severity.

Quantity	m	d	r	Mean	SD
Drug-response component	10.986	5	0.42		8
Placebo-response component	6.506	5	0.35		10
Natural-history component	6.506	5	0.35		10
Biomarker B_i				124.33	15.36
Baseline severity				83.07	18.48

4 Simulation design

We report the study in the ADEMP structure of Morris et al.[31].

Aims. To compare OL+BDC, CO and the Hybrid design on the power to detect $\beta_{bm:D}$, its bias and the calibration of its test, without dropout and under six dropout conditions, at one parameter point, with paired comparisons across conditions.

Data-generating mechanism. Data were generated by the package function `generateData()` under the mean-shift encoding (`dgp_architecture = 'mean_moderation'`, `moderation_scaling = 'constant'`). The response and baseline parameters are those of the published code of Hendrickson et al.[2], at commit 58b32a9, as stored in the package data. Table 1 lists them. The biomarker B_i has mean $\mu_{bm} = 124.33$ and standard deviation $\sigma_{bm} = 15.36$. Its standardized value is $b_i = (B_i - \mu_{bm})/\sigma_{bm}$. At every on-drug visit, the drug-response component is shifted by $\beta_{bm} b_i \sigma_{BR}$ with $\beta_{bm} = 0.45$ and $\sigma_{BR} = 8$. Off-drug visits receive no shift, so the shift has no carryover. The mean drug response itself carries over after discontinuation with half-life $t_{1/2} = 0.5$ week, decaying from the last on-drug mean. The three response components follow modified Gompertz means $G(t_{od}; m, d, r)$, the curve $m \exp\{-d \exp(-rt)\}$ shifted vertically to pass through the origin, with maximum m, displacement d and rate r. Each component has first-order autoregressive correlation $\rho = 0.7$ per week. Components are correlated at 0.1 at the same visit and 0.05 across visits. Every covariance matrix was positive definite, and the driver refuses any matrix that is not.

The components reduce symptoms, so a larger drug response lowers Y_{it}. The interaction is therefore negative. The analysis enters the standardized biomarker b_i. The true value is $\beta_{bm:D} = -\beta_{bm}\sigma_{BR} = -3.60$ points per biomarker standard deviation, under binary drug exposure. Under the null regime $\beta_{bm} = 0$ and the true value is 0.

Every design has a total of $N = 70$ participants, allocated across its paths: 35 on each path of OL+BDC and CO, and 18, 18, 17 and 17 on the four Hybrid paths.

Pairing. For each design and replicate, one complete dataset and one matrix of uniform draws U_{it} were generated. Each of the seven conditions was applied to that dataset with those draws. The conditions therefore differ only in the hazard. The null regime used its own seeds and was paired across conditions in the same way. Datasets were not shared across designs, so between-design comparisons are unpaired.

Estimand. $\beta_{bm:D}$ under the hypothetical strategy, with true value -3.60 (alternative) or 0 (null).

Methods. Each post-dropout dataset was analyzed with

$$Y_{it} = \beta_0 + \beta_B b_i + \beta_T w_t + \beta_D D_{bc,it} + \beta_{bm:D} b_i D_{bc,it} + \beta_\eta \eta_t + u_i + \varepsilon_{it},$$

with a participant random intercept u_i and CAR(1) residual correlation in calendar weeks, fitted by restricted maximum likelihood with `nlme::lme`. Here b_i is the biomarker standardized by μ_{bm} and σ_{bm}. The map from B_i to b_i is affine, so it leaves the tests unchanged and expresses $\beta_{bm:D}$ per biomarker standard deviation. The design expectancy η_t enters for OL+BDC and the Hybrid design. CO has a constant post-baseline expectancy, so the term is omitted there, as in `lme_analysis()`. The exposure indicator $D_{bc,it}$ is 1 on drug, 0 before the first exposure, and $2^{-t_{sd}/t^a_{1/2}}$ off drug after exposure, where t_{sd} is the time since discontinuation. We used two analysis half-lives:

- $t^a_{1/2} = 0.5$ week (analysis `ta0.5`): the historical specification, equal to the half-life of the mean drug response. It reproduces `lme_analysis()` exactly. We checked this on one dataset per design and obtained identical coefficients, standard errors and p-values.
- $t^a_{1/2} = 0$ (analysis `ta0`): binary drug exposure. This matches the generating interaction, which has no carryover.

Each fit was tested two ways. The model-based test uses the `nlme` standard error and degrees of freedom. The CR2 test uses the bias-reduced cluster-robust variance estimator[32] with Satterthwaite degrees of freedom (`clubSandwich::coef_test`), as in the project's carryover-sensitivity driver. If the CAR(1) fit failed, the driver refitted without the residual correlation and recorded the fallback.

Performance measures. For each design, condition, analysis and test we report the rejection rate at $\alpha = 0.05$ (power under the alternative, size under the null), bias against the true value, the empirical standard deviation of $\hat{\beta}_{bm:D}$, the calibration ratio κ and the coverage of the 95 percent confidence interval. κ is the mean standard error divided by the empirical standard deviation of the estimates, and $\kappa > 1$ indicates a conservative test. Against the no-dropout condition on the same datasets we report the paired power loss, the paired shift in $\hat{\beta}_{bm:D}$ and the mean ratio of standard errors. Each carries its paired MCSE, the standard deviation of the per-replicate difference divided by the square root of the number of replicates. The MCSE of a rejection rate is $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$.

Replicates and seeds. The alternative regime has 1,000 replicates per design and the null regime 1,000. The seed of each design and replicate is `set.seed(P09_SEED + 100000 * d + r)` with `P09_SEED` equal to 20261029, where d is the design index (1 OL+BDC, 2 CO, 3 Hybrid) and r the replicate. The null regime adds 50,000. Replicates ran in parallel with `parallel::mclapply` on 8 workers, with single-threaded linear algebra.

Table 2: Dropout achieved under the alternative, mean over replicates. "Obs. lost": fraction of post-baseline observations deleted. "At off-drug visit": fraction of dropouts whose first missing visit was off drug. The correlations are between the indicator of ever dropping out and the participant's mean improvement from baseline over the complete schedule, or the biomarker (MCSE of each correlation at most 0.004).

Design	Condition	Ever dropped	Obs. lost	At off-drug visit	Cor. with improvement	C
OL+BDC	MCAR 25%	0.252	0.148	0.278	-0.001	(
OL+BDC	Magnitude 25%	0.253	0.149	0.241	0.131	(
OL+BDC	Magnitude 40%	0.402	0.244	0.221	0.170	(
OL+BDC	Worsening 25%	0.251	0.150	0.270	-0.284	-
OL+BDC	Improvement 25%	0.252	0.150	0.269	0.285	(
OL+BDC	Symptom return 25%	0.252	0.147	0.284	-0.024	(
CO	MCAR 25%	0.252	0.148	0.497	0.002	-
CO	Magnitude 25%	0.250	0.140	0.452	0.109	(
CO	Magnitude 40%	0.402	0.232	0.450	0.139	(
CO	Worsening 25%	0.251	0.148	0.504	-0.255	-
CO	Improvement 25%	0.252	0.148	0.501	0.266	(
CO	Symptom return 25%	0.250	0.146	0.502	-0.024	-
Hybrid	MCAR 25%	0.249	0.146	0.419	0.002	(
Hybrid	Magnitude 25%	0.250	0.147	0.386	0.137	(
Hybrid	Magnitude 40%	0.400	0.243	0.366	0.170	(
Hybrid	Worsening 25%	0.249	0.146	0.409	-0.279	-
Hybrid	Improvement 25%	0.251	0.148	0.406	0.277	(
Hybrid	Symptom return 25%	0.246	0.144	0.420	-0.012	(

5 Results

Every fit converged (100 percent of fits in every cell). Of 84,000 fits, 7 fell back to the model without the CAR(1) residual, all in OL+BDC. No cell had more than 0.2 percent fallback fits.

5.1 Dropout achieved and its direction

We first checked where dropout fell and whom it removed. Table 2 gives the results for the alternative regime. The fraction of participants who ever dropped out matched its target, as the calibration guarantees. The fraction of post-baseline observations lost was smaller, about 0.14 to 0.15 at the 25 percent target and 0.23 to 0.24 at 40 percent.

Where dropout fell differed by design. Dropouts occurred at an off-drug visit in 0.22 to 0.29 of cases in OL+BDC, 0.36 to 0.42 in the Hybrid design and 0.45 to 0.50 in CO. These fractions follow the share of off-drug visits in each design.

The direction of selection matched the construction of each hazard. Under the magnitude-of-change hazard, participants who dropped out were more improved than those who stayed. The correlation with mean improvement was 0.109 to 0.170. Every response component lowers the

Table 3: Performance without dropout. Bias (MCSE) against the true value −3.60. Emp. SD: empirical standard deviation of the estimates. MB: model-based test. CR2: cluster-robust test with Satterthwaite degrees of freedom. Cover: coverage of the 95 percent interval. Size: rejection rate in the null regime. MCSE of power at most 0.014; of size at most 0.008.

Design	$t^a_{1/2}$	Bias	Emp. SD	κ MB	κ CR2	Cover MB	Cover CR2	Power MB	Power CR2	Size MB	Size CR2
OL+BDC	0.5	-0.344 (0.040)	1.258	1.12	0.98	0.969	0.942	0.820	0.845	0.038	0.066
OL+BDC	0	0.045 (0.036)	1.139	1.09	0.97	0.972	0.941	0.831	0.835	0.043	0.062
CO	0.5	-0.012 (0.044)	1.383	0.93	0.97	0.936	0.947	0.795	0.736	0.057	0.043
CO	0	0.005 (0.044)	1.376	0.93	0.97	0.934	0.944	0.795	0.737	0.057	0.041
Hybrid	0.5	-0.229 (0.039)	1.239	1.00	0.96	0.956	0.948	0.860	0.837	0.036	0.046
Hybrid	0	0.059 (0.035)	1.120	1.00	0.96	0.953	0.951	0.871	0.853	0.042	0.047

symptom score, so the mean change from baseline is an improvement at every post-baseline visit. A large squared change is therefore usually a large improvement. The worsening-driven and improvement-driven hazards gave correlations near -0.27 and 0.28. The symptom-return hazard was nearly unrelated to mean improvement (-0.024 to -0.012). Under the alternative, improvement-driven dropout removed high-biomarker participants more often (correlation 0.060 to 0.077), because those participants had the larger drug response. Worsening-driven dropout did the reverse (correlation -0.073 to -0.054). The MCSE of each correlation was at most 0.004.

Because α was calibrated on each path separately, every path lost the same expected fraction of participants. The per-path fractions stored by the driver confirm this. A comparison of dropout shares across the Hybrid paths, the quantity a path decomposition would need, is therefore fixed by construction in this study and carries no information.

5.2 Bias, calibration and the analysis half-life

We next compared the two analysis half-lives without dropout. Table 3 gives the results. With the historical half-life of 0.5 week, the interaction estimate was biased by -0.344 (0.040) in OL+BDC and -0.229 (0.039) in the Hybrid design, on a true value of −3.60. These are 10 and 6 percent of the true value, away from zero. The cause is the mismatch between the analysis and the generating process. The analysis gives off-drug visits after discontinuation an exposure weight of $2^{-t_{sd}/0.5}$, while the generating shift is zero at those visits. The bias in CO was -0.012 (0.044). CO has few off-drug visits soon after a discontinuation, because its visits are 2.5 weeks apart. With a binary exposure indicator, the bias was 0.045 (0.036) in OL+BDC, 0.059 (0.035) in the Hybrid design and 0.005 (0.044) in CO. Each of these was within 1.7 MCSE of zero.

The mismatch did not change power materially. The historical analysis inflated both the estimate and its standard error. Its empirical standard deviation in OL+BDC was 1.258 against 1.139 for the binary analysis. Model-based power was 0.820 and 0.831. CO had the lowest power under both half-lives and both tests. The Hybrid design and OL+BDC differed in power by at most 0.040, and their order depended on the test and the half-life. From here on we report the binary analysis, which is unbiased for this generating process.

The two tests were calibrated differently. The model-based test was conservative in OL+BDC.

Table 4: Power (MCSE) of the binary analysis ($t^a_{1/2} = 0$) by design, test and dropout condition, 1,000 paired datasets per design. MB: model-based. CR2: cluster-robust with Satterthwaite degrees of freedom.

Design	Test	None	MCAR 25%	Magnitude 25%	Magnitude 40%	Worsening 25%	Improvement 25%	Symptom return 25%
OL+BDC	MB	0.831 (0.012)	0.782 (0.013)	0.776 (0.013)	0.738 (0.014)	0.759 (0.014)	0.762 (0.013)	0.776 (0.013)
CO	MB	0.795 (0.013)	0.723 (0.014)	0.727 (0.014)	0.662 (0.015)	0.704 (0.014)	0.696 (0.015)	0.712 (0.014)
Hybrid	MB	0.871 (0.011)	0.831 (0.012)	0.829 (0.012)	0.797 (0.013)	0.803 (0.013)	0.825 (0.012)	0.825 (0.012)
OL+BDC	CR2	0.835 (0.012)	0.802 (0.013)	0.806 (0.013)	0.757 (0.014)	0.785 (0.013)	0.795 (0.013)	0.788 (0.013)
CO	CR2	0.737 (0.014)	0.669 (0.015)	0.661 (0.015)	0.594 (0.016)	0.640 (0.015)	0.633 (0.015)	0.648 (0.015)
Hybrid	CR2	0.853 (0.011)	0.804 (0.013)	0.811 (0.012)	0.770 (0.013)	0.799 (0.013)	0.801 (0.013)	0.809 (0.012)

Its κ under the alternative was 1.09, and its null rejection rate in this run was 0.043 (0.006). It was close to nominal in the Hybrid design, with κ 1.00 and null rejection rate 0.042 (0.006). In CO its κ was 0.93 and its null rejection rate 0.057 (0.007). Without dropout the CR2 test rejected in 0.041 (0.006) (CO), 0.047 (0.007) (Hybrid) and 0.062 (0.008) (OL+BDC) of the 1,000 null datasets of this run. Each rate was within 1.6 MCSE of nominal.

The same driver was applied to further independent null datasets (`analysis/scripts/cr2-olbdc-check/`). In OL+BDC, with 5,000 datasets at $N = 70$, the CR2 test rejected in 0.051 (0.003), and with 5,000 datasets at $N = 35$ in 0.049 (0.003). Its standard error was 0.98 times the empirical standard deviation at $N = 70$. The model-based test was conservative there, rejecting in 0.035 (0.003). With 1,000 datasets each, the CR2 test rejected in 0.057 (0.007) in the Hybrid design and 0.050 (0.007) in CO. The model-based test rejected in 0.049 (0.007) and 0.064 (0.008). In CO the model-based rate was 1.8 MCSE above nominal.

The seven dropout conditions of this run reuse the same 1,000 null datasets per design, so their null rejection rates are not independent replications. Across all 21 cells the model-based test rejected in 0.031 to 0.057 and the CR2 test in 0.038 to 0.065. Comparisons of power across designs under the model-based test therefore mix information with test calibration. For that reason we also compare designs by the empirical standard deviation of the estimate, which does not depend on the test (Section 5.5).

Dropout did not change the size of either test. Across the six dropout conditions, the paired change in the null rejection rate ranged from -0.012 to 0.009, with paired MCSE at most 0.008.

5.3 Power by design and dropout condition

Table 4 gives power for the binary analysis under both tests. Hybrid and OL+BDC had model-based power of 0.871 and 0.831 without dropout. CO had 0.795. CO had the lowest power in every condition under both tests. Section 5.5 compares the Hybrid design with OL+BDC.

5.4 Paired losses, coefficient shifts and standard-error ratios

Table 5 gives the paired comparisons against the no-dropout condition on the same datasets. Pairing reduced the MCSE of a power loss by a factor of about 1.8 relative to independent cells.

Table 5: Paired comparison of each dropout condition with no dropout, binary analysis. Loss: power without dropout minus power with it (paired MCSE). Shift: mean change in the estimate (paired MCSE). A positive shift moves the estimate toward zero. SE ratio: mean ratio of model-based standard errors (paired MCSE). SD ratio: ratio of empirical standard deviations.

Design	Condition	Loss MB	Loss CR2	Shift	SE ratio	SD ratio
OL+BDC	MCAR 25%	0.049 (0.009)	0.033 (0.009)	-0.039 (0.015)	1.089 (0.001)	1.054
OL+BDC	Magnitude 25%	0.055 (0.009)	0.029 (0.010)	-0.024 (0.016)	1.087 (0.001)	1.042
OL+BDC	Magnitude 40%	0.093 (0.011)	0.078 (0.012)	0.021 (0.021)	1.154 (0.002)	1.089
OL+BDC	Worsening 25%	0.072 (0.010)	0.050 (0.010)	0.057 (0.016)	1.077 (0.001)	1.066
OL+BDC	Improvement 25%	0.069 (0.010)	0.040 (0.010)	0.106 (0.017)	1.079 (0.001)	1.021
OL+BDC	Symptom return 25%	0.055 (0.010)	0.047 (0.010)	0.074 (0.017)	1.072 (0.001)	1.075
CO	MCAR 25%	0.072 (0.011)	0.068 (0.011)	0.004 (0.018)	1.079 (0.001)	1.074
CO	Magnitude 25%	0.068 (0.011)	0.076 (0.012)	0.063 (0.019)	1.065 (0.001)	1.052
CO	Magnitude 40%	0.133 (0.013)	0.143 (0.013)	0.142 (0.024)	1.118 (0.002)	1.091
CO	Worsening 25%	0.091 (0.011)	0.097 (0.012)	0.093 (0.019)	1.060 (0.001)	1.071
CO	Improvement 25%	0.099 (0.012)	0.104 (0.012)	0.204 (0.021)	1.060 (0.001)	1.060
CO	Symptom return 25%	0.083 (0.011)	0.089 (0.012)	0.094 (0.020)	1.069 (0.001)	1.044
Hybrid	MCAR 25%	0.040 (0.008)	0.049 (0.009)	-0.005 (0.014)	1.081 (0.001)	1.057
Hybrid	Magnitude 25%	0.042 (0.008)	0.042 (0.008)	0.028 (0.014)	1.076 (0.001)	1.049
Hybrid	Magnitude 40%	0.074 (0.010)	0.083 (0.011)	0.036 (0.020)	1.136 (0.002)	1.110
Hybrid	Worsening 25%	0.068 (0.010)	0.054 (0.009)	0.064 (0.015)	1.073 (0.001)	1.066
Hybrid	Improvement 25%	0.046 (0.009)	0.052 (0.009)	0.072 (0.016)	1.071 (0.001)	1.073
Hybrid	Symptom return 25%	0.046 (0.009)	0.044 (0.009)	0.063 (0.015)	1.067 (0.001)	1.047

Three results follow from the table.

First, the rate of dropout set the loss of precision, and the design did not. At 25 percent MCAR dropout the standard error rose by a factor of 1.079 to 1.089 across designs. At 40 percent magnitude-driven dropout it rose by 1.118 to 1.154. The paired MCSE of each ratio was at most 0.002. The 25 percent conditions of different mechanisms gave ratios within 0.029 of each other.

Second, the mechanism changed the estimate. MCAR dropout moved it by at most 0.039 (paired MCSE at most 0.018). The three directional mechanisms moved it toward zero in every design, each by at least 3.4 paired MCSE. Improvement-driven dropout moved it by 0.204 (0.021) in CO, 0.106 (0.017) in OL+BDC and 0.072 (0.016) in the Hybrid design. Worsening-driven dropout moved it by 0.057 to 0.093, and symptom-return dropout by 0.063 to 0.094. Magnitude-driven dropout moved the CO estimate toward zero by 0.142 (0.024) at 40 percent. In OL+BDC and the Hybrid design it moved the estimate by at most 0.036. With the binary exposure indicator, the shift in each outcome-dependent condition was largest in CO. With the 0.5-week indicator, OL+BDC shifted more than CO under symptom-return dropout (0.110 against 0.093) and by a similar amount under worsening-driven dropout (0.096 against 0.095). Improvement-driven dropout removes high-biomarker participants, who have the largest drug response (Section 5.1). The observed data then understate the interaction. Worsening-driven dropout removes low-biomarker participants and also attenuated the estimate. We did not isolate the cause of that attenuation. The largest shift, 0.204 in CO, is 5.7 percent of the true value. Coverage of the 95 percent interval stayed between 0.936 and 0.959 for the CR2 test in every cell, because the shifts were small relative to the standard error.

Third, CO lost more absolute power than the other two designs in every condition. At 40 percent magnitude-driven dropout, model-based power fell by 0.133 (0.013) in CO, 0.093

(0.011) in OL+BDC and 0.074 (0.010) in the Hybrid design. The difference between CO and the Hybrid design was 0.059 (0.017) ($z = 3.56$). Across the six dropout conditions, the CO loss exceeded the Hybrid loss by more than two standard errors in 5 of 6 conditions under the CR2 test and 4 of 6 under the model-based test. Against OL+BDC the counts were 6 and 1.

The starting power does not explain this difference. A normal approximation predicts the loss from the no-dropout power π_0 and the standard-error ratio s as $\pi_0 - \Phi\{(\Phi^{-1}(\pi_0) + 1.96)/s - 1.96\}$. Adding the paired shift Δ multiplies $\Phi^{-1}(\pi_0) + 1.96$ by $1 - \Delta/|\beta_{bm:D}|$.

For 40 percent magnitude-driven dropout the first form gives 0.093 for CO, 0.116 for OL+BDC and 0.094 for the Hybrid design. The observed losses were 0.133, 0.093 and 0.074. The approximation predicts nearly equal losses from the starting power, while the observed CO loss was the largest. With the shift added, it gives 0.128 for CO, close to the observed loss. It then gives 0.121 and 0.102 for OL+BDC and the Hybrid design, above their observed losses. The extra loss in CO coincides with the attenuation of its estimate. The approximation does not account for all of the differences between designs, and we did not resolve the remainder.

5.5 Design comparison on precision

Without dropout, the empirical standard deviation of the binary estimate was 1.120 in the Hybrid design, 1.139 in OL+BDC and 1.376 in CO (MCSE at most 0.031). The relative efficiency of CO to the Hybrid design was $(1.120/1.376)^2 = 0.66$. CO would need about 1.5 times as many participants to estimate the interaction as precisely, if precision scales with N. The Hybrid and OL+BDC designs differed by 2 percent in standard deviation. Their power losses differed by at most 0.023 across conditions and tests. The largest $|z|$ for a difference in their losses was 1.73 among the 12 comparisons, which is within the range expected from chance alone.

5.6 Comparison with the earlier unpaired prototype

An earlier version of this study used independent datasets for each cell, 500 replicates per cell, the historical analysis and the magnitude-of-change hazard under the label ‘biased’. It used an earlier parameter set that did not match the published code, not that of Table 1. The prototype was not rerun. It reported power losses at 40 percent of 0.086 (CO), 0.058 (Hybrid) and 0.056 (OL+BDC), with MCSE 0.015 to 0.025. The paired driver, run under that earlier parameter set with the same analysis, gave 0.111 (0.011), 0.058 (0.008) and 0.065 (0.009) (`analysis/data/informative-dropout-by-design/`). The two runs agree within their Monte Carlo error. Under the parameters of Table 1 the paired run with the same analysis gave 0.134 (0.013), 0.083 (0.011) and 0.108 (0.012). The earlier run could not separate the CO loss from the others. The paired run separates it.

The earlier run also reported an OL row. That row tested the biomarker-by-time slope (Section 3), not the interaction, and is not carried forward.

Table 6: What the paired cells support, binary analysis unless stated, model-based shifts. Shifts are toward zero.

Situation	Finding
No dropout or MCAR dropout	Hybrid and OL+BDC equal in precision. CO needs about 1.5 times the sample
Dropout driven by the size of symptom change	Standard error rises by a similar factor in all designs (1.118 to 1.154 at 40 percent). CO estimate attenuated by 0.142 at 40 percent
Worsening-driven dropout	Estimate attenuated by 0.057 to 0.093; precision loss as for MCAR
Improvement-driven dropout	Estimate attenuated by 0.072 (Hybrid) to 0.204 (CO)
Dropout on symptom return	Estimate attenuated by 0.063 to 0.094
Choice of analysis half-life	Mismatch biased OL+BDC by -0.34 and Hybrid by -0.23

6 Implications for design choice

Table 6 states what the simulated cells support. It applies to this parameter point only.

On these results the Hybrid design and OL+BDC are interchangeable for the interaction. The choice between them can rest on participant burden and feasibility. CO is less efficient for this estimand at this parameter point. Its estimate was the most sensitive to every outcome-dependent dropout mechanism, and most of all to improvement-driven dropout. The analysis half-life should match the generating interaction. In practice the true form is unknown, and a data-chosen half-life is the project's provisional recommendation (paper 02). We did not evaluate it here.

7 Discussion

7.1 What the paired design changed

Applying every dropout condition to the same datasets turned each loss and shift into a paired difference. This made the CO loss separable from the others. On the standard-error scale the three designs lost the same precision to dropout of a given rate. CO still lost more absolute power. Its estimate was also attenuated most by outcome-dependent dropout, and the extra loss coincides with that attenuation. The earlier unpaired prototype had read its 2.8-point difference in losses between CO and the Hybrid design as evidence that CO is the least robust design. With a standard error of about 2.9 points, that difference was not resolved.

7.2 Why the results differ from Hendrickson et al.

Hendrickson et al.[2] found the Hybrid design slightly less powerful than CO without carryover. Here CO was the least efficient design. The two studies share the response and baseline parameters of Table 1. They differ in several other ways that could explain the difference. The original DGP uses compound symmetry with a common within-component correlation of 0.8. We used first-order autoregressive correlation of 0.7 per week. The original DGP encodes

the interaction as a correlation between the biomarker and the drug response. We used the mean-shift encoding. The present carryover half-life is 0.5 week. Hendrickson et al. report that even 0.1 week of carryover reduced the power of the Hybrid design sharply. Their dropout hazard is per unit time and built on the shifted squared change, and ours is per visit. Which of these accounts for the difference was not tested.

7.3 Limitations

- **One parameter point.** Sample size, effect size, carryover half-life and correlation parameters were fixed. The protocol's sweeps over $N \in \{35, 70, 100\}$ and $\beta_{bm} \in \{0, 0.3, 0.6\}$ were not run.
- **One encoding.** The interaction was generated by the mean-shift encoding only. Under the correlation encoding the interaction has a different exposure profile, and the results may differ.
- **No shared-parameter dropout.** The hazards depend on observed outcomes. A hazard tied to the latent drug response would be more directly informative about the biomarker.
- **No analysis-stage sensitivity.** Every condition was analyzed with the likelihood-based mixed model under an implicit MAR assumption. Pattern-mixture, delta-adjusted and reference-based analyses[11,12,20–22] and tipping-point displays[25] were not applied.
- **No path decomposition.** The dropout calibration fixes each path's dropout fraction, so the Hybrid paths cannot be compared on dropout share. A path-stratified analysis would need a calibration across the whole design.
- **Per-visit hazard.** Visit spacing differs by design, so equal targets do not mean equal hazards per week.
- **Calibrated rates.** The ever-dropout fraction is set by construction. The study compares mechanisms at equal rates. It does not predict how often participants will leave a given design.
- **Test calibration.** The model-based test was conservative in OL+BDC and above nominal in CO. Power comparisons across designs under that test carry this difference. The null regime has 1,000 datasets per design, so single-cell null rejection rates carry an MCSE of about 0.007. The CR2 calibration rests on separate checks of 5,000 datasets in OL+BDC and 1,000 in each of the other two designs.
- **Platform.** The run was made in a Linux container (Ubuntu 24.04.4 LTS), not in a host R session. Results are not draw-for-draw reproducible across platforms.
- **One parameter set.** The response and baseline parameters are those of the published code. Results at other response means and variances may differ.
- **Symptom-return mechanism.** Its hazard uses the step change between consecutive visits, which differ in spacing. It was only weakly related to mean improvement and is one of several possible operationalizations.

8 Conclusions

At one parameter point with $N = 70$, the Hybrid and OL+BDC designs estimated the biomarker-by-treatment interaction with the same precision and lost the same share of it to dropout. CO needed about 1.5 times the sample for the same precision. It also lost more absolute power under dropout. Its lower starting power did not explain that loss. The attenuation of its estimate under outcome-dependent dropout accounts for part of it.

The rate of dropout set the loss of precision. The mechanism set the bias. Improvement-driven dropout attenuated the estimate by up to 5.7 percent of its true value. Worsening-driven and symptom-return dropout attenuated it by up to 2.6 percent. Magnitude-driven dropout attenuated it by up to 4.0 percent, in CO. MCAR dropout moved it by at most 1.1 percent. Dropout did not change test size.

An analysis half-life of 0.5 week biased the OL+BDC and Hybrid estimates by about 10 and 6 percent, because the generating interaction had no carryover. The analysis must match the exposure profile of the interaction, not only that of the mean drug response.

Glossary

- **Analysis `ta0.5`.** The historical analysis: the exposure indicator decays after discontinuation with analysis half-life $t^a_{1/2} = 0.5$ week. It reproduces the package function `lme_analysis()`.
- **Analysis `ta0`.** The same model with $t^a_{1/2} = 0$, so the exposure indicator is binary. It matches the generating interaction.
- **Calibration ratio κ.** The mean standard error divided by the empirical standard deviation of the estimates. Values above 1 indicate a conservative test.
- **CR2.** The bias-reduced cluster-robust variance estimator of Bell and McCaffrey, clustered on participant, with Satterthwaite degrees of freedom.
- **CO.** The traditional two-period crossover: four visits on one treatment, then four on the other, in randomized order.
- **Dropout conditions.** `none`; `MCAR_25` (constant hazard, 25 percent ever drop out); `magnitude_25` and `magnitude_40` (hazard rising in the squared, uncentered, scaled change from baseline; stored in the earlier prototype as `biased_25` and `biased_40`); `worsening_25` and `improvement_25` (hazard rising or falling in the centered, scaled change from baseline); `return_25` (hazard rising in the centered, scaled change since the previous visit). Section 2 gives the formulas.
- **Hybrid design.** Open-label titration, then blinded discontinuation, then a brief two-visit crossover, with four randomization paths[2].
- **Mean-shift encoding.** The data-generating form in which the biomarker shifts the drug-response mean at on-drug visits by $\beta_{bm} b_i \sigma_{BR}$ (package value `dgp_architecture = 'mean_moderation'`).
- **OL, OL+BDC.** Open label throughout; open-label titration followed by blinded discontinuation.

- **Paired loss.** The mean over replicates of the rejection indicator without dropout minus the rejection indicator with dropout, computed on the same datasets.
- **Parameter set.** The response and baseline parameters of the simulation. This run uses the set of the published code of Hendrickson et al. (Table 1, data label `published`). The earlier prototype used an earlier set.
- **Analysis half-life $t^a_{1/2}$.** The half-life the analysis uses to build the exposure indicator. It is distinct from the data-generating half-life $t_{1/2}$.

Abbreviations

CAR(1), continuous-time first-order autoregressive; CO, crossover; CR2, bias-reduced cluster-robust variance estimator; MAR, missing at random; MCAR, missing completely at random; MCSE, Monte Carlo standard error; MNAR, missing not at random; OL, open label; OL+BDC, open label plus blinded discontinuation; SE, standard error.

Conflict of interest

The authors declare no conflicts of interest.

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Data availability statement

The data and code that support the findings of this study are in the pmsimstats-ng project repository. Per-replicate outputs and summary files are under `analysis/data/informative-dropout-by-desig` The drivers are under `analysis/scripts/informative-dropout-by-design/`. Exact paths are listed in the Reproducibility section.

Reproducibility

The paired study is produced by `analysis/scripts/informative-dropout-by-design/20-paired-dropout.R` with `--params published`, run from the repository root through `PMSIM_PARAMS=published bash analysis/scripts/informative-dropout-by-design/20-run-paired.sh`. The wrapper sets `VECLIB_MAXIMUM_THREADS`, `OPENBLAS_NUM_THREADS` and `OMP_NUM_THREADS` to 1. The environment variables `P09_REPS` and `P09_NULL_REPS` set the replicate counts (this manuscript used 1,000 and 1,000). The driver writes `paired-replicates.rds` (one row per design, regime, replicate, condition and analysis), `paired-dropout.rds`

(dropout diagnostics) and `paired-meta.rds` (settings, parameter set, elapsed time and session information) to `analysis/data/informative-dropout-by-design-published/`. The script `21-summarise-paired.R --params published` in the same directory writes the four summary files `paired-cells.csv`, `paired-loss.csv`, `paired-contrast.csv` and `paired-dropout.csv`. The null checks of the tests are produced by `analysis/scripts/cr2-olbdc-check/` under the same parameter set and summarized in `analysis/data/cr2-olbdc-check-published/summary.csv`. Every number in Sections 5 to 8 is read from those files by inline R when this manuscript is rendered.

Seeds are `set.seed(P09_SEED + 100000 * d + r)` for design index d and replicate r, plus 50,000 in the null regime, with `P09_SEED` equal to 20261029. The run reported here was made in a Linux container (R version 4.6.0 (2026-04-24), Ubuntu 24.04.4 LTS). Results from a host R session are not draw-for-draw identical because the linear-algebra libraries differ. The run took 59 minutes on 8 workers.

The earlier unpaired prototype is `analysis/scripts/quick-sim/09-dropout-prototype-10min.R`. It calls `set.seed(20260508)` once and seeds each replicate with `100000 * i + rep` for cell index i. It hard-codes the earlier parameter set. Its outputs are `analysis/data/quick-sim/09-dropout-replica` and `09-dropout-summary.txt`. The working protocol `analysis/scripts/informative-dropout-by-design/` and its scaffold drivers (`00-common.R`, `01-study1.R` to `04-study4.R`) were not run.

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